SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

12. Perform Perspective Transformation on the image.

AIM:
Perspective Transformation on the image.

PROGRAM:
```
import cv2
import numpy as np

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if img is None:
    print("Error: Image not found!")
else:
    rows, cols = img.shape[:2]

    src_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, rows - 1],
        [cols - 1, rows - 1]
    ])

    dst_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [int(0.33 * cols), rows - 1],
        [int(0.66 * cols), rows - 1]
    ])

    M = cv2.getPerspectiveTransform(src_points, dst_points)

    perspective_img = cv2.warpPerspective(img, M, (cols, rows))

    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Perspective_Transformed_Image.jpg',
        perspective_img
```

```
)

cv2.imshow("Perspective Transformed Image", perspective_img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:

OUTPUT:

RESULT:

Thus the program is executed successfully.